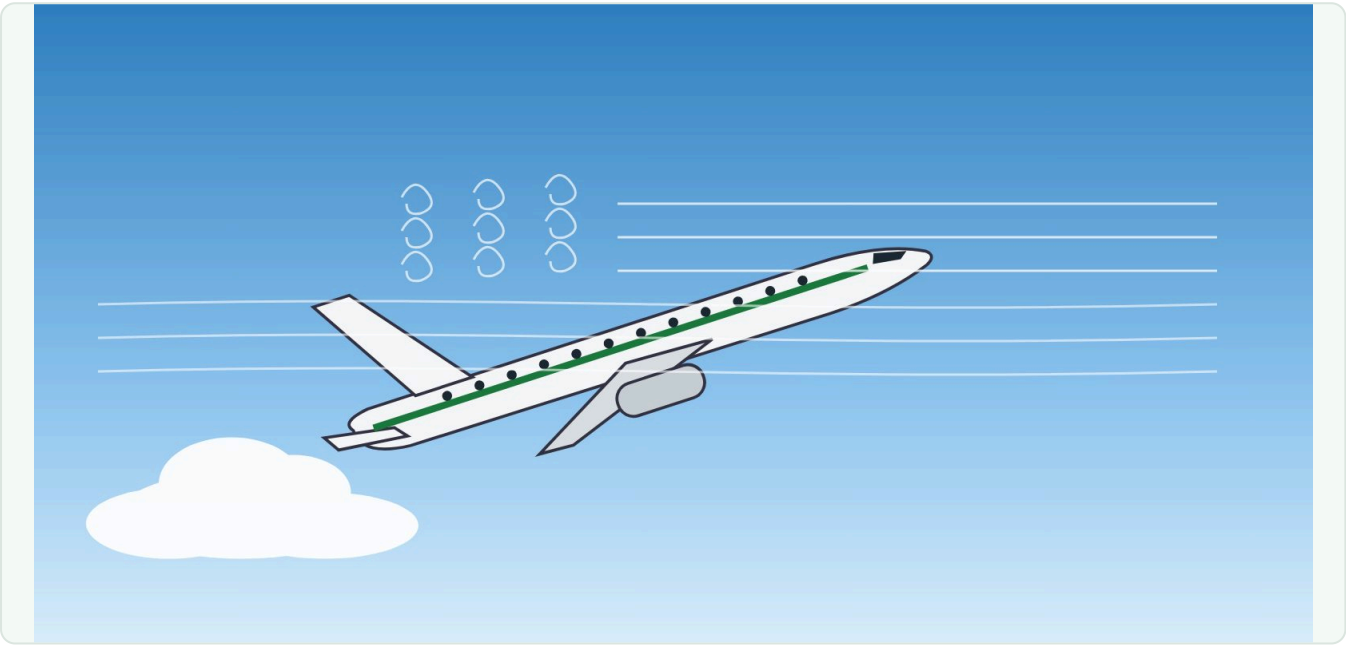
MODULE 2 · AERODYNAMICS

Lesson 05: The Stall

Lesson 5 of 24 · about 9 min

LEARNING OBJECTIVES

- Describe the aerodynamic characteristics of an approach to stall and a full stall.
- Identify stall warning systems and natural stall cues.
- Explain the factors that change stall speed and stall behaviour.



In a stall the airflow separates from the upper surface of the wing.

Approach to stall and full stall

- **Approach to stall (impending stall)** — the angle of attack is approaching the critical value. Indications: stall warning (stick shaker, aural warning), low-speed indications, and possibly the onset of buffet.
- **Full stall** — the critical angle of attack has been exceeded. Indications may include buffet (sometimes heavy), an uncommanded nose-down pitch or roll, inability to arrest a descent, loss of pitch authority, and stick pusher activation on aircraft that have one.

Both require the same immediate action: **reduce the angle of attack**.

Stall warning and protection systems

- **Stall warning** (stick shaker, aural alert) activates with a margin before the stall — on transport aeroplanes, at a speed and angle of attack set by certification requirements.
- **Stick pusher** (some types) pushes the control column forward to prevent the aircraft from reaching an aerodynamic stall. Do not fight it.

- **Envelope protection** (fly-by-wire types in normal law) limits angle of attack — but protections can be lost when the control law degrades, for example with unreliable air data.
- Stall warning is scheduled for a **clean wing**. With ice or contamination, the stall may occur **before** the warning.

What changes the stall speed

Factor	Effect on stall speed
Higher weight	Increases (stall speed is proportional to √weight)
Higher load factor (turns, pull-ups)	Increases with √n — about 41 % higher at 2 g (Lesson 6)
Flaps and slats extended	Decreases (higher CLmax)
Ice or contamination	Increases, and the critical angle of attack decreases
Forward centre of gravity	Increases slightly (more tail down-force needed)
Higher altitude	Indicated stall speed changes little at low altitude, but Mach effects raise it at high altitude (Lesson 7)

Stall behaviour

- Swept wings tend to stall at the tips first, which can cause roll-off and a nose-up pitching tendency. Design features (washout, vortex generators, fences) reduce this.
- Aircraft with T-tails may be susceptible to a **deep stall**, which is why they are often fitted with stick pushers.
- Power effects: on aircraft with underwing engines, high thrust gives a nose-up pitching moment (Lesson 10) which can work against stall recovery.
- Simulators may not reproduce full-stall behaviour accurately outside their validated envelope (Lesson 24).

> **REMEMBER**
> Pulling back to "hold altitude" when the stall warning activates increases the angle of attack — the opposite of what is needed. Altitude loss is secondary; **reducing the angle of attack is the priority**.

KEY TAKEAWAYS

- Approach to stall and full stall both require an immediate reduction in angle of attack.
- Stall warning is scheduled for a clean wing — ice can cause a stall without warning.
- Weight, load factor, configuration and contamination change the stall speed.
- Do not fight a stick pusher.

KNOWLEDGE CHECK

The stick shaker activates during a level-off. What is the first priority?

A. Maintain altitude with back pressure

B. Reduce the angle of attack

C. Increase bank to turn away

D. Extend the speedbrakes

Answer: B. The stall warning indicates a high angle of attack. Reducing the angle of attack is the first priority, even at the cost of altitude.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.